Lean formalization of an Assumptions of Physics technical brief: log

This log records which statements of the technical brief in this directory were checked in Lean, where the formal version differs from the brief, and any problems found in the brief. The brief is available at `https://assumptionsofphysics.org/resources/briefs`.

Setup

- Lean `v4.36.0-rc1`; Mathlib at commit `2cabef97`; physlib (`master`) and TauCeti (commit `eb4dd24`, on the same Mathlib) as further dependencies.
- Build: `lake build` in the project root.
- Every file is checked with `#print axioms` on its main results. A result is recorded here only if it depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`, and so has no `sorry`.
- Imports are kept to the Mathlib modules actually needed.
- Material that belongs in Mathlib is kept in `ForMathlib/`, in Mathlib's directory layout (library `ForMathlib`), and imported by the brief files.

1 Non-linear representations in quantum mechanics

Files.

- File: `briefs/NonLinearQuantum/NonLinearQuantum.lean`
- Library: `NonLinearQuantum`
- Imports: `Mathlib.Analysis.InnerProductSpace.Calculus`, `Mathlib.Analysis.SpecialFunctions.ExpDeriv`

Setting. $H = \mathbb{C}^2$ (`EuclideanSpace ℂ (Fin 2)`), with $|z^+\rangle = e_0$, $|z^-\rangle = e_1$, and

$$m(v) = e^{i\theta\, p(z^+|v)}\, v, \qquad p(z^+ \mid v) = \frac{|v_0|^2}{\|v\|^2}.$$

In Lean $m(0) = 0$ holds without a case split, since $0/0 = 0$.

Results.

Brief	Lean
eq. (3), rays to rays	`nlMap_smul`: $m(kv) = k\,m(v)$ for every $k \in \mathbb{C}$
each ray is mapped to itself	`nlMap_mem_ray`
eq. (4), m not linear	`nlMap_not_add`, `not_isLinearMap_nlMap`, for $\theta \in (0, 2\pi)$
eq. (5), inner product	`inner_nlMap`
eq. (6), Born rule preserved	`born_nlMap`
Remark: m bijective	`nlHomeomorph`: a homeomorphism of $\mathbb{C}^2$, inverse $m_{-\theta}$
Remark: m a diffeomorphism (correction)	`contDiffAt_nlMap`: smooth at every $v \neq 0$
	`not_differentiableAt_nlMap_zero`: not differentiable at 0
Theorem 1, mathematical content	`exists_nonlinear_born_preserving`
Born rule depends only on rays	`born_smul`: $p(cv \mid dw) = p(v \mid w)$ for $c, d \neq 0$
$m(H)$, structure carried along m	`Twisted`, `toTwisted` (identity of the carrier)
same scalars, hence same rays	`toTwisted_smul`
same Born rule	`born_toTwisted`
different addition	`not_map_add_toTwisted`
(correction) $H \cong m(H)$	`twistedIsometry`: m is a linear isometry onto $m(H)$
Theorem 1, precise form	`exists_second_linear_structure`
(caveat) dimension one	`map_add_of_map_smul`: maps commuting with scalars are additive

Differences from the brief. Theorem 1 ("REP-VSLIN is independent of QST") is a statement about the logical independence of two conditions, which the brief does not make precise. Two things are checked.

- The existence of a homeomorphism of H that maps every ray to itself, preserves the Born rule and is not additive.
- A precise form of ¬REP-VSLIN under which the brief's proof is a proof: the linear structure of H is not fixed by the rays and the Born rule. The type `Twisted` θ is the carrier of H with addition $v \oplus w = m(m^{-1}v + m^{-1}w)$, scalar multiplication $k \odot v = m(k\,m^{-1}v)$ and inner product $\langle m^{-1}v, m^{-1}w\rangle$; it is a complex inner product space. By eq. (3) its scalar multiplication is that of H, so it has the same rays; by eq. (6) it has the same Born rule; by eq. (4) its addition differs from that of H.

See the open problems for why the brief has to choose this reading.

Problems found in the brief.

- The closing remark says that m is a diffeomorphism. It is not differentiable at 0: along every real line $m(tv) = t\,m(v)$, so a derivative at 0 would have to equal m itself, and m is not additive. The correct statement is that m is a homeomorphism of $\mathbb{C}^2$ and a diffeomorphism of $\mathbb{C}^2 \setminus \{0\}$. Theorem 1 only needs a bijection, so it is not affected.
- The proof says that "since the map is non-linear, $H \not\equiv m(H)$ in the category of vector spaces". This is false. As a set $m(H) = H$, since m is a bijection. With the vector space structure

of H, $m(H)$ is H. With the structure carried over along m, m is by construction a linear isometry $H \to m(H)$ (`twistedIsometry`). What is true is that the identity of the carrier is not linear from one structure to the other.

- Eq. (6) holds for every map that sends each vector into its own ray, since the Born rule only sees rays (`born_smul`). Any $v \mapsto \varphi([v])\, v$ with $\varphi\colon P(H) \to \mathbb{C} \setminus \{0\}$ would do. What matters for the proof is that m commutes with scalars (eq. (3)) and is not additive (eq. (4)).

- QST allows $\dim H \leq 1$. There every map that commutes with scalars is additive (`map_add_of_map_smul`), so there is no second linear structure. Theorem 1 claims only consistency, so the two-state example suffices; "for every system" would need $\dim H \geq 2$.